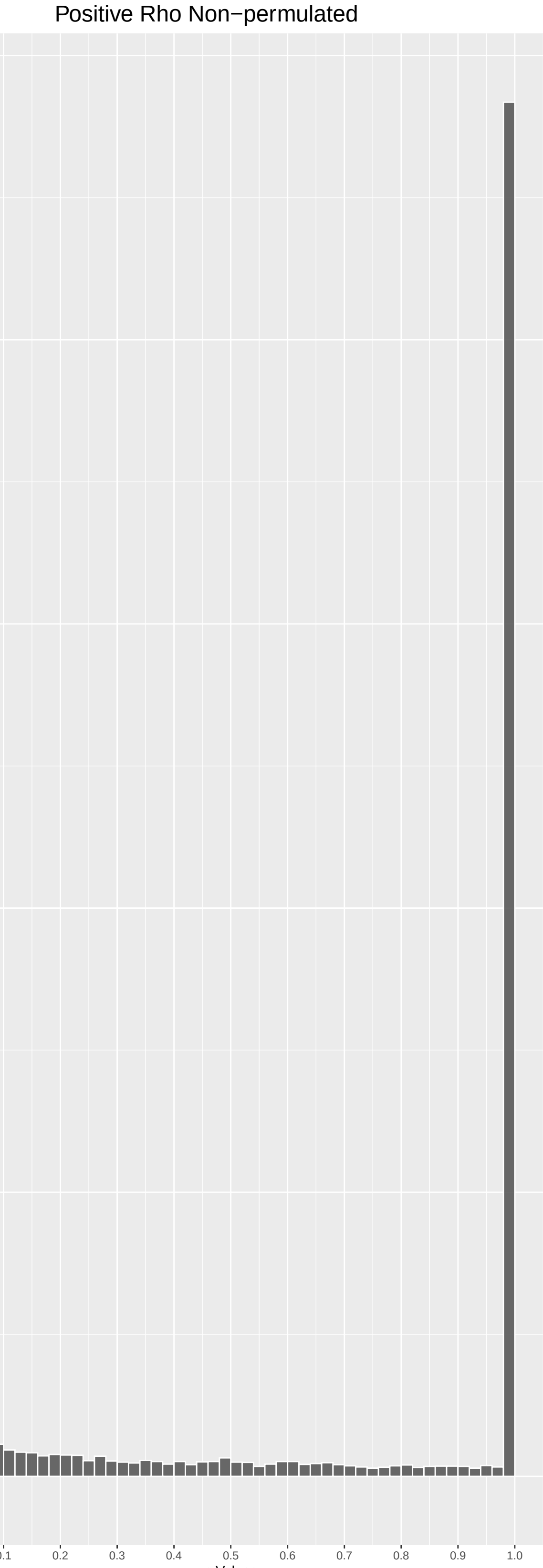


Top genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
EP400	7.306784	1.641666e-12	2.631e-08	2.646e-04
TGFBRAP1	7.182008	4.121701e-12	3.303e-08	2.646e-04
MUC20	-6.998354	1.553923e-11	8.301e-08	4.434e-04
HERC2	6.702683	1.227766e-10	4.919e-07	1.817e-03
NSD2	6.604507	2.393060e-10	6.801e-07	1.817e-03
NEK5	-6.595306	2.546285e-10	6.801e-07	1.817e-03
TRPM5	6.271608	2.144023e-09	4.909e-06	1.050e-02
LRRC53	-6.226888	2.854755e-09	5.719e-06	1.050e-02
PDE6B	6.203618	3.310781e-09	5.895e-06	1.050e-02
ADH6	-5.992855	1.237129e-08	1.836e-05	2.676e-02
ADGRB1	5.989808	1.260534e-08	1.836e-05	2.676e-02
AKAP9	-5.943985	1.669049e-08	2.229e-05	2.694e-02
NEDD4	-5.909721	2.056126e-08	2.354e-05	2.694e-02
SHANK2	5.920197	1.929337e-08	2.354e-05	2.694e-02
RIMBP2	5.798697	4.009927e-08	4.284e-05	4.239e-02
CHST15	5.783652	4.385756e-08	4.393e-05	4.239e-02
PKD1L3	-5.691512	7.555155e-08	7.122e-05	4.239e-02
ZSCAN20	-5.677036	8.222921e-08	7.321e-05	4.239e-02
CYP4X1	5.663997	8.873212e-08	7.484e-05	4.239e-02
LRPPRC	-5.650471	9.600519e-08	7.519e-05	4.239e-02
SLBP	5.617819	1.160300e-07	7.519e-05	4.239e-02
KCNK18	5.615947	1.172933e-07	7.519e-05	4.239e-02
NAV1	5.633852	1.057368e-07	7.519e-05	4.239e-02
PIGG	5.631783	1.070134e-07	7.519e-05	4.239e-02
IRAG2	-5.629246	1.085995e-07	7.519e-05	4.239e-02

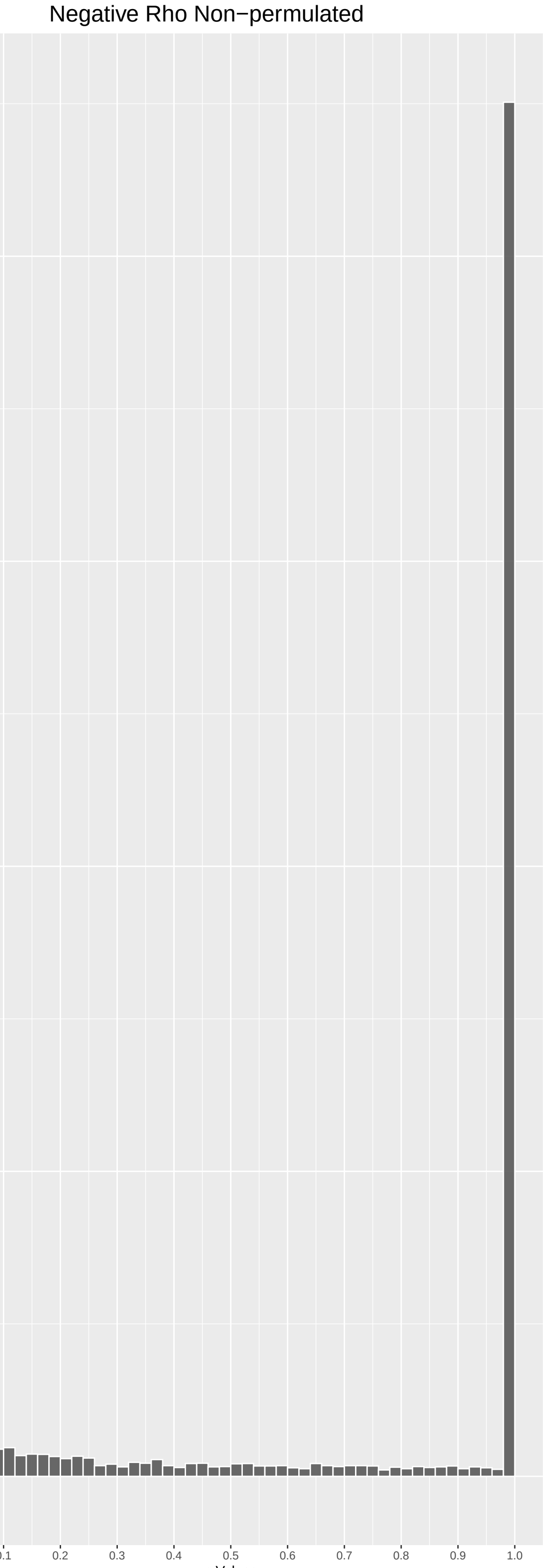


Top Positive genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
EP400	7.306784	1.641666e-12	2.631e-08	2.646e-04
TGFBRAP1	7.182008	4.121701e-12	3.303e-08	2.646e-04
HERC2	6.702683	1.227766e-10	4.919e-07	1.817e-03
NSD2	6.604507	2.393060e-10	6.801e-07	1.817e-03
TRPM5	6.271608	2.144023e-09	4.909e-06	1.050e-02
PDE6B	6.203618	3.310781e-09	5.895e-06	1.050e-02
ADGRB1	5.989808	1.260534e-08	1.836e-05	2.676e-02
SHANK2	5.920197	1.929337e-08	2.354e-05	2.694e-02
RIMBP2	5.798697	4.009927e-08	4.284e-05	4.239e-02
CHST15	5.783652	4.385756e-08	4.393e-05	4.239e-02

Top genes by Q-Value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
EP400	7.306784	1.641666e-12	2.631e-08	2.646e-04
TGFBRAP1	7.182008	4.121701e-12	3.303e-08	2.646e-04
MUC20	-6.998354	1.553923e-11	8.301e-08	4.434e-04
HERC2	6.702683	1.227766e-10	4.919e-07	1.817e-03
NSD2	6.604507	2.393060e-10	6.801e-07	1.817e-03
NEK5	-6.595306	2.546285e-10	6.801e-07	1.817e-03
PDE6B	6.203618	3.310781e-09	5.895e-06	1.050e-02
LRRC53	-6.226888	2.854755e-09	5.719e-06	1.050e-02
TRPM5	6.271608	2.144023e-09	4.909e-06	1.050e-02
ADH6	-5.992855	1.237129e-08	1.836e-05	2.676e-02
ADGRB1	5.989808	1.260534e-08	1.836e-05	2.676e-02
AKAP9	-5.943985	1.669049e-08	2.229e-05	2.694e-02
NEDD4	-5.909721	2.056126e-08	2.354e-05	2.694e-02
SHANK2	5.920197	1.929337e-08	2.354e-05	2.694e-02
USP11	5.586753	1.388128e-07	7.671e-05	4.239e-02
LRPPRC	-5.650471	9.600519e-08	7.519e-05	4.239e-02
ANKRD12	-5.589323	1.367736e-07	7.671e-05	4.239e-02
RGS9	5.603297	1.261876e-07	7.671e-05	4.239e-02
SLBP	5.617819	1.160300e-07	7.519e-05	4.239e-02
KCNK18	5.615947	1.172933e-07	7.519e-05	4.239e-02
ZSCAN20	-5.677036	8.222921e-08	7.321e-05	4.239e-02
NAV1	5.633852	1.057368e-07	7.519e-05	4.239e-02
CYP4X1	5.663997	8.873212e-08	7.484e-05	4.239e-02
CHST15	5.783652	4.385756e-08	4.393e-05	4.239e-02
PIGG	5.631783	1.070134e-07	7.519e-05	4.239e-02



Top Negative genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
MUC20	-6.998354	1.553923e-11	8.301e-08	4.434e-04
NEK5	-6.595306	2.546285e-10	6.801e-07	1.817e-03
LRRC53	-6.226888	2.854755e-09	5.719e-06	1.050e-02
ADH6	-5.992855	1.237129e-08	1.836e-05	2.676e-02
AKAP9	-5.943985	1.669049e-08	2.229e-05	2.694e-02
NEDD4	-5.909721	2.056126e-08	2.354e-05	2.694e-02
PKD1L3	-5.691512	7.555155e-08	7.122e-05	4.239e-02
ZSCAN20	-5.677036	8.222921e-08	7.321e-05	4.239e-02
LRPPRC	-5.650471	9.600519e-08	7.519e-05	4.239e-02
IRAG2	-5.629246	1.085995e-07	7.519e-05	4.239e-02

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NADH Dehydrogenase Complex Assembly (GO:000072)	-0.25444780	48	1.096e-09	1.844e-06	TMEM126A:53 NDUF88:76 NDUF53:118 TMEM126B:156 ECSIT:201 NDUFAF1:400
Chemical Synaptic Transmission (GO:00072)	0.11153131	254	1.098e-09	1.184e-06	RIMBP2:9 GABRB3:19 CACNA1B:45 NPTX1:62 APBA2:108 GABRG3:113
Mitochondrial Gene Expression (GO:014005)	-0.17737969	100	9.372e-10	1.184e-06	POLRMT:50 PTCD3:74 FASTKD5:208 FASTKD1:315 DAP3:377 GFM2:439
Mitochondrial Respiratory Chain Complex	-0.25444780	48	1.096e-09	1.184e-06	TMEM126A:53 NDUF88:76 NDUF53:118 TMEM126B:156 ECSIT:201 NDUFAF1:400
Monoatomic Cation Transmembrane Transport	0.10801384	273	9.703e-10	1.184e-06	TRPM5:5 KCNK18:15 TRPM1:53 TMEM175:93 SCN2A:109 SLC6A15:120
Mitochondrial Translation (GO:0032543)	-0.17698092	96	2.183e-09	1.960e-06	PTCD3:74 DAP3:377 GFM2:439 MRPS23:446 MRPL3:501 TSFM:616
Potassium Ion Transmembrane Transport (G	0.15037876	132	2.631e-09	2.025e-06	KCNK18:15 TMEM175:93 LETM1:112 KCNA10:139 KCNJ12:201 KCNQ3:205
tRNA Modification (GO:0006400)	-0.20377625	66	1.069e-08	7.200e-06	THADA:83 FTSJ1:146 DTVD1:148 TRMT10A:203 ALKBH8:244 PUS10:368
Inorganic Cation Transmembrane Transport	0.09875084	277	1.805e-08	1.080e-05	TRPM5:5 KCNK18:15 TRPM1:53 TMEM175:93 SCN2A:109 SLC6A15:120
Potassium Ion Transport (GO:0006813)	0.14857617	119	2.304e-08	1.241e-05	KCNK18:15 TMEM175:93 KCNA10:139 KCNJ12:201 KCNQ3:205 KCNF1:208
tRNA Methylation (GO:0030488)	-0.26821183	36	2.600e-08	1.274e-05	THADA:83 FTSJ1:146 TRMT10A:203 METTL1:383 THUMP02:411 TRMT10B:426
Metal Ion Transport (GO:0030001)	0.11891711	157	2.914e-07	1.242e-04	KCNK18:15 ITPR1:89 SCN2A:109 TRPV6:138 KCNA10:139 KCNJ12:201
Mitochondrial Respiratory Chain Complex	-0.16908301	77	2.997e-07	1.242e-04	TMEM126A:53 NDUF88:76 NDUF53:118 TMEM126B:156 ECSIT:201 NDUFAF1:400
DNA Repair (GO:0006281)	-0.08929724	273	4.310e-07	1.659e-04	BRCA1:15 FANCM:27 BRCA2:29 MLH3:47 MBD4:82 DCLRE1A:87
Inorganic Cation Import Across Plasma Me	0.13890582	96	2.649e-06	9.516e-04	CACNA1B:45 TRPM1:53 SCN2A:109 TRPV6:138 CACNA1I:150 KCNJ12:201
Calcium Ion Transport (GO:0006816)	0.12922299	106	4.454e-06	1.333e-03	TRPM5:5 TRPM1:53 ITPR1:89 TRPV6:138 TRPV2:272 RAMP3:273
Double-Strand Break Repair Via Homologou	-0.13050115	105	3.976e-06	1.333e-03	BRCA1:15 FANCM:27 BRCA2:29 R21:128 RNF169:161 HELQ:168
Visual Perception (GO:0007601)	0.13672208	95	4.255e-06	1.333e-03	TRPM1:53 CRYGA:98 VAX2:226 OAT:231 CRYBA2:235 RD3:291
Proton Motive Force-Driven Mitochondrial	-0.19011282	47	6.598e-06	1.871e-03	NDUF88:76 NDUF53:118 NDUF41:551 NDUF9A:555 NDUF85:710 ATP5PB:731
Calcium Ion Import Across Plasma Membran	0.22558774	33	7.359e-06	1.983e-03	CACNA1B:45 TRPM1:53 SCN2A:109 TRPV6:138 CACNA1I:150 TRPV2:272
DNA Metabolic Process (GO:0006259)	-0.07851654	274	8.490e-06	2.178e-03	BRCA1:15 MBD4:82 POLQ:103 RMI1:128 DNA2:151 STON1:184
RNA Methylation (GO:0001510)	-0.16960521	57	9.624e-06	2.357e-03	THADA:83 FTSJ1:146 TRMT10A:203 METTL4:290 METTL1:383 THUMP02:411
Oxidative Phosphorylation (GO:0006119)	-0.17201825	55	1.037e-05	2.429e-03	NDUF88:76 NDUF53:118 NDUF41:551 NDUF9A:555 NDUF85:710 ATP5PB:731
Proton Motive Force-Driven ATP Synthetis	-0.17715474	51	1.223e-05	2.745e-03	NDUF88:76 NDUF53:118 NDUF41:551 NDUF9A:555 NDUF85:710 ATP5PB:731
Cardiac Conduction (GO:0061337)	0.19081449	43	1.513e-05	3.135e-03	BIN1:206 KCNJ3:648 CAV1:776 DSG2:865 ANK2:898 DSC2:913
Double-Strand Break Repair (GO:0006302)	-0.10334891	157	1.540e-05	3.135e-03	BRCA1:15 BRCA2:29 DCLRE1A:87 POLQ:103 RMI1:128 DNA2:151
Regulation Of Heart Rate By Cardiac Cond	0.20216653	38	1.630e-05	3.253e-03	BIN1:206 KCNJ3:648 CAV1:776 DSG2:865 ANK2:898 DSC2:913
Mitochondrial Electron Transport, NADH T	-0.22026719	31	2.206e-05	4.246e-03	NDUF88:76 NDUF53:118 NDUF41:400 NDUF9A:555 NDUF85:710 NDUFV3:784
Sensory Perception Of Light Stimulus (GO	0.12495115	96	2.394e-05	4.448e-03	TRPM1:53 CRYGA:98 VAX2:226 OAT:231 CRYBA2:235 RD3:291
Aerobic Respiration (GO:0009060)	-0.16579436	54	2.541e-05	4.563e-03	NDUF88:76 NDUF53:118 MTFR2:278 NDUF41:551 NDUF9A:555 NDUF85:710
Mitochondrial RNA Metabolic Process (GO)	-0.27847400	19	2.652e-05	4.608e-03	POLRMT:50 FASTKD5:208 FASTKD1:315 FTB2M:455 PNPT1:456 MTERF2:1298
Muscle Contraction (GO:0006936)	0.13025599	87	2.743e-05	4.618e-03	MYH2:148 KCNJ12:201 LMOD1:245 EMD:280 LMOD2:283 MYH1:361
Modulation Of Chemical Synaptic Transmis	0.11161580	117	3.148e-05	5.140e-03	CACNA1B:45 NPTX1:62 MAPK8IP2:102 CPLX3:149 GRD1:183 TRIO:211
Adenylate Cyclase-Inhibiting G Protein-C	0.17557938	46	3.831e-05	5.756e-03	GNAI3:236 OPR1L:238 MCHR1:322 GPR37:1370 OPRK1:837 HRH3:858
Anterograde Trans-Synaptic Signaling (GO	0.08098616	181	3.764e-05	5.756e-03	GABRB3:19 CACNA1B:45 NPTX1:62 APBA2:108 KCNQ3:205 GLRA3:251
Sister Chromatid Segregation (GO:0000819)	-0.20714522	33	3.846e-05	5.756e-03	SPAG5:114 KIFC1:124 ZWINT:160 ZWINT:160 ZWINT:160 KIF18B:752
Cell Junction Assembly (GO:0034329)	0.13082932	82	4.308e-05	6.273e-03	SHANK2:8 SDK1:17 SDK2:64 SHANK3:134 PKP1:199 DBNL:393
Cell-Cell Junction Organization (GO:0045	0.14304803	67	5.231e-05	7.417e-03	PKP1:199 TGFBI:418 NLGN4X:591 FTRL1:599 GJA4:630 CDH6:676
Recombinational Repair (GO:0000725)	-0.12174079	89	7.365e-05	1.017e-02	BRCA1:15 BRCA2:29 RMI1:128 RNF169:161 HELQ:168 STON1:184
Striated Muscle Contraction (GO:0006941)	0.15453876	52	1.171e-04	1.577e-02	TNNT3:493 RYR2:636 NOS1:990 MYH7B:1043 TNNT2:1067 GRK2:1140

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
BENPORATH_ES_WITH_H3K27ME3	0.08238917	978	5.540e-18	3.594e-14	RG59:16 FRMPD4:30 CACNA1B:45 COL12A1:49 FZD10:60 NPTX1:62
MEISSNER_NPC_HCP_WITH_H3K4ME2_AND_H3K27M	0.13223583	330	1.827e-16	5.928e-13	ADGBR1:7 CADM3:80 STUM:81 GJA3:86 GABRG3:113 PTGFRN:147
MIKKELSEN_MEF_HCP_WITH_H3K27ME3	0.10080254	553	7.334e-16	1.586e-12	ADGBR1:7 CPZ:43 CDCP1:65 CADM3:80 MAPK8IP2:102 CNTNA:104
REACTOME_NEURONAL_SYSTEM	0.11521652	382	1.312e-14	2.128e-11	SHANK2:8 KCNK18:15 GABRB3:19 CACNA1B:45 ADCY1:68 APBA2:108
MIKKELSEN_MCV6_HCP_WITH_H3K27ME3	0.10893489	419	2.537e-14	3.292e-11	ADGBR1:7 FZD10:60 SDK2:64 STUM:81 LCAM:92 GRID2IP:100
NIKOLSKY_BREAST_CANCER_7P22_AMPLICON1	0.33879155	38	4.990e-13	4.625e-10	TTYH3:118 INTS1:129 ADAP1:165 TMEM184A:223 GRIFIN:263 SNX8:289
FISCHER_DREAM_TARGETS	-0.07277445	873	4.568e-13	4.625e-10	BRCA1:15 ATAD5:16 FAM161A:17 FANCM:27 BRCA2:29 ARL13B:37
BENPORATH_EED_TARGETS	0.07000951	920	9.596e-13	7.782e-10	FRMPD4:30 CACNA1B:45 COL12A1:49 FZD10:60 NPTX1:62 PARP12:63
MIKKELSEN_NPC_HCP_WITH_H3K27ME3	-0.12905611	329	2.235e-12	1.611e-09	ADGBR1:7 CPZ:43 CADM3:80 GABRG3:113 ADRA2C:160 C1QL2:186
MARSON_BOUND_BY_E2F4_UNSTIMULATED	-0.05811569	645	6.538e-11	4.242e-08	FANCM:27 BRCA2:29 DLGAP5:38 KIF24:52 KIAA0825:56 BUB1B:64
MARTENS_TRETINOIN_RESPONSE_UP	0.07255241	675	1.691e-10	9.973e-08	SHANK2:8 CHST15:10 FAM50B:42 CPZ:43 TRIM29:57 FZD10:60
BENPORATH_SUZ12_TARGETS	0.06253765	914	2.084e-10	1.927e-07	CYP4X1:11 FRMPD4:30 CACNA1B:45 FZD10:60 NPTX1:62 CADM3:80
JOHNSTONE_PARV_B_TARGETS_3_DN	-0.06584236	771	6.618e-10	3.303e-07	N4BP2:22 SRFBP1:31 DLGAP5:38 MYSL1:59 MIS18BP1:65 RTKN2:77
WP_MITOCHONDRIAL_COMPLEX_I_ASSEMBLY_MODE	-0.25349653	47	1.844e-09	7.977e-07	NDUF88:76 NDUF53:118 TMEM126B:156 ECSIT:201 NDUFAF1:400 NDUF4F2:407
HAMAI_APOPTOSIS_VIA_TRAIL_UP	-0.07280334	591	1.811e-09	7.977e-07	GOLGA4:13 BRCA1:15 DLGAP5:38 CWF19L2:51 FPGT:54 AGGF1:55
REACTOME_COMPLEX_I_BIOGENESIS	-0.24320429	48	5.618e-09	2.278e-06	NDUF88:76 NDUF53:118 TMEM126B:156 ECSIT:201 NDUFAF1:400 NDUF4F2:407
MEISSNER_BRAIN_HCP_WITH_H3K4ME3_AND_H3K2	0.05430730	1019	6.123e-09	2.337e-06	ELFN1:32 COL12A1:49 FZD10:60 INAVA:79 GRID2IP:100 VWCV2:111
BENPORATH_PRC2_TARGETS	0.07087314	577	7.147e-09	2.576e-06	FRMPD4:30 CACNA1B:45 FZD10:60 NPTX1:62 GRID1:183 PKP1:199
REACTOME_SIGNALING_BY_GPCR	0.06696149	626	1.299e-08	4.436e-06	RG59:16 FZD10:60 ADCY1:68 ITPR1:89 OPN3:91 ADORA1:133
RODRIGUES_THYROID_CARCINOMA_POORLY_DIFFE	-0.06847274	588	1.677e-08	5.440e-06	NEDD4:6 LRPPRC:9 BRCA1:15 DLGAP5:38 AGGF1:55 BUB1B:64
REACTOME_DEVELOPMENTAL_BIOLOGY	0.05513236	909	2.295e-08	7.092e-06	DSG3:52 LCAM:92 KLK13:105 SCN2A:109 MMP2:128 KRT80:144
REACTOME_MITOCHONDRIAL_TRANSLATION	-0.16792014	92	2.655e-08	7.830e-06	PTCD3:74 DAP3:377 GFM2:439 MRPS23:446 MRPL3:501 TSFM:616
REACTOME_MUSCLE_CONTRACTION	0.11812164	185	3.160e-08	8.914e-06	KCNK18:15 NEB:38 MYBP2:44 ITPR1:89 SCN2A:109 CACNA1I:150
KEGG_NEUROACTIVE_LIGAND_RECEPTOR_INTERAC	0.09955384	261	3.302e-08	8.927e-06	GABRB3:19 GABRG3:113 ADORA1:133 MASI:135 ADRA2C:160 MC4R:174
PEREZ_TP53_TARGETS	0.05119429	1022	4.232e-08	1.098e-05	TGFBRA1:2 RIMBP2:9 CHST15:10 PHRF1:35 SIPA1L2:55 INSYN2A:66
REACTOME_KERATINIZATION	0.17180569	85	4.452e-08	1.111e-05	DSG3:52 KLK13:105 KRT80:144 TGM5:179 PKP1:199 KRT5:216
BLANCO_MELO_BRONCHIAL_EPITHELIAL_CELLS_1	-0.12140502	163	9.237e-08	2.220e-05	IRAG2:10 BRCA1:15 ATAD5:16 FAM161A:17 KIF24:52 BUB1B:64
GEORGES_TARGETS_OF_MIR192_AND_MIR215	-0.05528389	810	1.113e-07	2.578e-05	BRCA1:15 ATAD5:16 N4BP2:22 FANCM:27 BRCA2:29 DLGAP5:38
REACTOME_POTASSIUM_CHANNELS	0.16153439	90	1.203e-07	2.691e-05	KCNK18:15 KCNA10:139 KCNJ12:201 KCNQ3:205 KCNF1:208 KCNQ2:399
DDDD_NASOPHARYNGEAL_CARCINOMA_DN	-0.04495147	1212	1.932e-07	4.158e-05	NEDD4:6 LRPPRC:9 BRCA1:15 DLGAP5:38 AGGF1:55 BUB1B:64
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT	-0.16349496	81	3.698e-07	7.739e-05	LRPPRC:9 NDUF88:76 NDUF53:118 TMEM126B:156 ECSIT:201 NDUFAF1:400
REACTOME_G_ALPHA_I_SIGNALLING_EVENTS	0.09021648	267	4.134e-07	8.382e-05	RG59:16 ADCY1:68 ITPR1:89 OPN3:91 ADORA1:133 TAS2R3:137
KAUFFMANN_DNA_REPAIR_GENES	-0.10115368	210	4.571e-07	8.986e-05	BRCA1:15 N4BP2:22 FANCM:27 BRCA2:29 MLH3:47 MBD4:82
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT	-0.14476455	101	5.078e-07	9.690e-05	LRPPRC:9 NDUF88:76 NDUF53:118 TMEM126B:156 ECSIT:201 NDUFAF1:400
MIKKELSEN_NPC_HCP_WITH_H3K4ME3_AND_H3K27	0.10254165	199	6.398e-07	1.186e-04	ELFN1:32 GJA3:86 MAPK8IP2:102 SLC6A15:120 ADORA1:133 PTGFRN:147
REACTOME_VOLTAGE_GATED_POTASSIUM_CHANNEL	0.22118240	42	7.097e-07	1.279e-04	KCNA10:139 KCNQ3:205 KCNF1:208 KCNQ2:399 KCNGA:404 KCNS2:439
REACTOME_GPCR_LIGAND_BINDING	0.07251855	398	7.524e-07	1.319e-04	FZD10:60 OPN3:91 ADORA1:133 TAS2R3:137 ADRA2C:160 MC4R:174
KEGG_CALCIIUM_SIGNALING_PATHWAY	0.10966575	171	7.798e-07	1.331e-04	CACNA1B:45 ADCY1:68 ITPR1:89 CACNA1I:150 P2RX2:212 ADRB1:304
FLORIO_NEOCORTEX_BASAL_RADIAL_GLIA_DN	-0.10679752	178	9.223e-07	1.534e-04	BRCA1:15 BRCA2:29 DLGAP5:38 BUB1B:64 MIS18BP1:65 BRP1:89
REACTOME_FORMATION_OF_THE_CORNIFIED_ENVE	0.16014563	78	1.023e-06	1.660e-04	DSG3:52 KLK13:105 KRT80:144 TGM5:179 PKP1:199 KRT5:216

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Schizophrenia	0.04441659	1613	7.628e-09	7.483e-05	HERC2:3 SHANK2:8 NAV1:12 RGS9:16 GABRB3:19 SORL1:24
Mitochondrial Diseases	-0.08320443	346	1.236e-07	6.061e-04	LRPPRC:9 TMEM126A:53 NDUF87:76 NDUF53:118 DNA2:151 TMEM126B:156
Nuclear cataract	0.23678770	32	3.585e-06	1.172e-02	GJA3:86 GJA8:125 CRYBA2:235 HSF4:366 CRYBB1:347 MIP:739
Bipolar Disorder	0.04904596	720	9.487e-06	1.861e-02	GABRB3:19 SORL1:24 TRANK1:41 CACNA1B:45 PER2:50 NPTX1:62
Primary microcephaly	-0.12630292	104	8.848e-06	1.861e-02	FANCM:27 BRCA2:29 NIN:80 BRP1:89 DNA2:151 TRMT10A:203
Nuclear non-senile cataract	0.23491970	29	1.202e-05	1.965e-02	GJA3:86 GJA8:125 CRYBA2:235 CRYBB1:347 MIP:739 BFPSP1:740
Autism Spectrum Disorders	0.05630441	481	2.713e-05	3.803e-02	HERC2:3 SHANK2:8 GABRB3:19 EHMT1:29 ITPR1:89 LCAM:92
Chromosome Breakage	-0.18648627	40	4.520e-05	5.543e-02	BRCA1:15 FANCM:27 BRCA2:29 BRP1:89 FANCF:246 ATM:308
Increased CSF lactate	-0.15939819	53	6.037e-05	6.580e-02	LRPPRC:9 NDUF53:118 TMEM126B:156 NDUFAP1:400 NDUF4F2:407 TRMT10C:441
CATARACT, COPPOCK-LIKE	0.34829421	10	1.369e-04	6.918e-02	GJA3:86 BFPSP2:116 GJA8:125 CRYBB1:347 CRYGC:1006 CRYGD:1443
Cataract, Puvulenter	0.36126761	10	7.628e-05	6.918e-02	GJA3:86 BFPSP2:116 GJA8:125 HSF4:366 CRYBB1:347 CRYGC:1006
Diabetes Mellitus, Experimental	0.10960651	101	1.444e-04	6.918e-02	SOD1:87 MMP2:128 PDK4:130 BAX:262 TGFBI:418 NOS3:430
Psychoses, Substance-Induced	0.27694438	16	1.256e-04	6.918e-02	RG59:16 ESR1:395 SOD2:1074 HTR1A:1390 CAMK2B:1444 PICK1:2193
Seizures, Focal	0.09996239	123	1.325e-04	6.918e-02	SCN2A:109 LETM1:112 KCNQ3:205 DYNC1H1:230 CLU:340 SLC6A2:349
Alloxan Diabetes	0.10960651	101	1.444e-04	6.918e-02	SOD1:87 MMP2:128 PDK4:130 BAX:262 TGFBI:418 NOS3:430
Hyperinsulinism	0.06228548	335	9.743e-05	6.918e-02	HERC2:3 PDE6B:6 EHMT1:29 IGFBP3:74 PDK4:130 MC4R:174
Lactic acidemia	-0.10159103	120	1.246e-04	6.918e-02	LRPPRC:9 NDUF53:118 TMEM126B:156 NDUFAP1:400 NDUF4F2:407 TRMT10C:441
Nicotinamide adenine dinucleotide coenzy	-0.18288058	39	7.811e-05	6.918e-02	NDUF53:118 NDUF4F1:400 NDUF4F2:407 NDUF41:551 NDUF412:753 HDH17B6:858
Severe myopia	0.11669994	91	1.218e-04	6.918e-02	PDE6B:6 TRPM1:53 IGFBP3:74 CRYBA4:394 ESR1:395 TGFBI:418
Sloping forehead	-0.12361814	79	1.481e-04	6.918e-02	KAT6B:176 CPT2:237 SETBP1:243 NHEJ1:31 CEP290:358 CENPE:395
Streptozotocin Diabetes	0.10960651	101	1.444e-04	6.918e-02	SOD1:87 MMP2:128 PDK4:130 BAX:262 TGFBI:418 NOS3:430
Alcoholic Intoxication, Chronic	0.05729313	373	1.590e-04	7.089e-02	GABRB3:19 ADCY1:68 CNTNA:104 GABRG3:113 MMP2:128 OPR1L:238
Alzheimer's Disease	0.02865683	1646	1.704e-04	7.269e-02	SHANK2:8 RNFL12:20 SORL1:24 EHMT1:29 TBC1D8:40 PER2:50
Acidosis, Lactic	-0.08853475	150	1.884e-04	7.393e-02	LRPPRC:9 NDUF53:118 TMEM126B:156 NDUFAP1:400 NDUF4F2:407 TRMT10C:441
HIV Infections	0.04423281	626	1.851e-04	7.393e-02	ENTPD1:33 TMEM132D:75 MMP2:128 NTSE1:312 FUT1:195 PPP1R1B:398
Calcanoevalgus deformity	0.39955410	7	2.514e-04	8.973e-02	TNNT3:439 COL1A2:974 TNNT1:2067 PHGDH:1752 EZH2:1892 MYH3:2150
Congenital cataract	0.11226976	89	2.561e-04	8.973e-02	GJA3:86 CRYGA:98 BFPSP2:116 GJA8:125 CRYBA2:235 HSF4:366
Mouth Neoplasms	0.10330837	106	2.434e-04	8.973e-02	PTK2:178 PTHFR:279 HMGCS2:282 UPT1:339 CDKN2A:91 CXCL12:643
Increased serum lactate	-0.10860622	95	2.781e-04	9.408e-02	LRPPRC:9 NDUF53:118 POLG:234 NDUF4F2:407 TRMT10C:441 NDUF49:555
Waddling gait	0.15122583	48	2.922e-04	9.554e-02	NEB:38 BICD2:39 SOD1:87 LCAM:92 BIN1:206 DYNC1H1:230
Leigh Disease	-0.11639064	80	3.246e-04	9.952e-02	LRPPRC:9 NDUF53:118 NDUF4F2:407 MSHE:453 NPNT1:45 NDUF49:555
Visual seizure	0.07580860	191	3.158e-04	9.952e-02	BRD1:85 LETM1:112 ARC:242 CLU:340 SLC6A2:349 CHRNA3:400
Status Epilepticus, Subclinical	0.12672994	65	4.152e-04	1.020e-01	PTK2B:351 GRM1:448 SSTR4:520 SLC12A5:522 ATP2A2:771 CDH2:948
Complex Partial Status Epilepticus	0.12672994	65	4.152e-04	1.020e-01	PTK2B:351 GRM1:448 SSTR4:520 SLC12A5:522 ATP2A2:771 CDH2:948
Grand Mal Status Epilepticus	0.12672994	65	4.152e-04	1.020e-01	PTK2B:351 GRM1:448 SSTR4:520 SLC12A5:522 ATP2A2:771 CDH2:948
Lipoprotein (a) measurement	0.28534145	13	3.681e-04	1.020e-01	SLC22A3:265 SYTL3:423 DSCAM1L1:542 SLC22A1:803 TAGAP:784 IGFBP2:1088
Non-Convulsive Status Epilepticus	0.12672994	65	4.152e-04	1.020e-01	PTK2B:351 GRM1:448 SSTR4:520 SLC12A5:522 ATP2A2:771 CDH2:948
Petit mal status	0.12120217	71	4.190e-04	1.020e-01	NOTCH1:185 PTK2B:351 GRM1:448 SSTR4:520 SLC12A5:522 ATP2A2:771
Pfaunder-Hurter Syndrome	-0.21226770	23	2.465e-04	1.020e-01	ARSB:113 VCAN:226 MAN2B1:458 NAGL:905 TBI1:987 GNPATB:998
Simple Partial Status Epilepticus	0.12672994	65	4.152e-04	1.020e-01	PTK2B:351 GRM1:448 SSTR4:520 SLC12A5:522 ATP2A2:771 CDH2:948